

Decentralised Ticketing System

CS 218- Programmable and Interoperable Blockchains

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Introduction

Event ticketing has always been an essential part of the entertainment and cultural industry. However, traditional ticketing platforms are riddled with inefficiencies, high costs, and risks. Users often face the threat of purchasing fake or duplicated tickets. Organizers, on the other hand, must deal with high platform fees, restricted control, and limited transparency over sales.

To address these issues, our project proposes a Decentralized Event Ticketing System that leverages the Ethereum blockchain and Non-Fungible Tokens (NFTs) to build a trustless, secure, and user-friendly alternative. The blockchain ensures immutability and transparency, while NFTs provide a digital, verifiable representation of each ticket that cannot be forged or duplicated. This system empowers users to verify, own, transfer, and even resell their tickets with full autonomy and security.

Problem Identification

2.1 Ticket Fraud and Scalping

Ticket fraud has become increasingly sophisticated. Scammers produce counterfeit tickets that look convincingly real, which are then sold to unsuspecting buyers. Upon reaching the event, the victims are denied entry due to invalid tickets.

Scalping refers to the practice where bots or resellers buy tickets in bulk as soon as they are released and resell them at extremely inflated prices. This not only makes it difficult for genuine attendees to purchase tickets but also damages the reputation of the organizers and limits the reach of the event.

2.2 High Intermediary Fees

Most traditional ticketing platforms operate as centralized entities. They charge significant service, transaction, and convenience fees to both organizers and customers. These fees reduce the profit margins of event hosts and increase the final ticket price for attendees. Organizers also face delays in receiving payments and lack control over the resale or transfer of tickets.

Proposed Solution

To solve the above problems, we propose an on-chain ticketing system that uses smart contracts to issue and manage tickets in the form of NFTs. These NFTs represent unique, verifiable, and programmable digital assets stored on the Ethereum blockchain.

3.1 *Use of Ethereum Blockchain*

The Ethereum blockchain provides several benefits:

- **Decentralization:** No single entity controls the system, preventing misuse.
- **Transparency:** All ticket transactions are recorded on-chain and can be publicly verified.
- **Immutability:** Once data is written to the blockchain, it cannot be altered or deleted.
- **Smart Contracts:** They automate all processes like minting, resale, and refunds, eliminating the need for middlemen.

3.2 *NFTs as Tickets*

Each ticket is minted as a unique ERC-721 token:

- **Minting:** Tickets are generated and stored as NFTs, giving each one a unique identity.
- **Ownership:** Only one user can hold a ticket at any given time. Ownership is easily verifiable.
- **Transferability:** Tickets can be securely transferred from one wallet to another, including resale scenarios, without relying on a central authority.

This mechanism reduces fraud, enables fair resale, and offers complete control to both buyers and organizers.

Workflow of NFT Ticketing System

The following workflow outlines the lifecycle of a ticket in our decentralized system:

1. *Event Creation and Ticket Minting:*

The organizer creates an event and specifies parameters such as name, location, date, number of tickets, and ticket price. The smart contract then mints the specified number of NFT tickets and makes them available for purchase.

2. *Ticket Purchase by Users:*

Users connect their Ethereum wallet (e.g., MetaMask) and purchase tickets by sending ETH to the smart contract. Upon successful payment, an NFT is minted and transferred to the buyer's wallet.

3. *Validation at Entry Point:*

At the event venue, the NFT ticket is validated by scanning the wallet address or QR code associated with it. Because NFTs are immutable and unique, this prevents duplication or forgery.

4. *Resale and Cancellation:*

If a user cannot attend, they may resell the ticket using the platform's resale function. The smart contract ensures authenticity and handles payment transfer. In case of cancellation, users can burn their NFT ticket and receive a 90% refund automatically through the smart contract.

Technical Implementation

5.1 Development Stack

- **Smart Contract Language:** Solidity
- **NFT Standard:** ERC-721
- **Deployment Tool:** Hardhat
- **Blockchain:** Ethereum Testnet (Goerli or Sepolia)
- **Testing Tools:** Mocha, Chai, and Hardhat console

5.2 Testing and Verification

We thoroughly tested the platform to ensure all use cases work as expected:

- Creation of events and validation of metadata.
- Minting of NFT tickets and correct seat allocation.
- Ensuring refund logic is executed accurately.
- Verification of transfer ownership during resale.
- Confirmation that events and transactions are recorded on-chain via Etherscan.

Smart Contract Architecture

The core functionality of our ticketing system is implemented through a **Solidity smart contract** that follows the **ERC-721 standard** for NFTs. This contract manages the full lifecycle of an event and its tickets in a decentralized, trustless manner.

Below are the key functions and their purposes:

- **createEvent()**
Used by the organizer to set up a new event. It includes details like event name, location, date, ticket price, and total seats. Once deployed, it enables ticket minting.
- **getOccasion()**
A public function to retrieve all event details. It helps users verify the event metadata before purchasing a ticket.
- **getSeatsTaken()**
Returns a list of seats that are already booked. This ensures users can choose from the remaining available seats and avoids double booking.
- **mint()**
Called when a user wants to buy a ticket. After verifying payment and seat availability, the contract mints a unique NFT and transfers it to the buyer's wallet.

- **buyTicket() / sellTicket()**

Enables resale between users. The owner can list their ticket, and another user can buy it securely. Ownership is automatically updated via the blockchain.

- **cancelTicket()**

Lets a user cancel their ticket if needed. The NFT is burned, and the user receives a 90% refund. This adds flexibility while discouraging misuse.

- **withdraw()**

Allows the event organizer to collect the funds from ticket sales. Only the organizer has permission to perform this action.

- **receive()**

A fallback function that safely accepts ETH sent directly to the contract without any function call. It ensures no funds are lost due to user error.

These functions work together to ensure secure event creation, ticket ownership, transfers, cancellations, and payments — all without relying on any centralized authority.

Deployment Summary

The decentralized ticketing system was successfully deployed on the Ethereum testnet via a smart contract named **TokenMaster**, operating under the token symbol **TM**. This contract manages the full lifecycle of NFT-based event ticketing using the **ERC-721** standard.

To test real-world functionality, we deployed several sample events—including *UFC Miami*, *ETH Tokyo*, and *NBA Match in Dallas*—covering various scenarios like ticket minting, resale, and cancellation.

Each event was listed on-chain, and all transactions were verified using Ethereum testnet explorers like Goerli or Sepolia Etherscan. This enabled transparent tracking of user actions such as event creation, ticket purchases, transfers, and refunds.

The deployment confirmed that:

- Events were correctly initialized and stored on-chain.
- NFT tickets were securely minted and transferred upon payment.
- Ticket resale and ownership transfers functioned reliably.
- Refunds and NFT burning during cancellations worked as expected.
- Organizers could withdraw collected funds without issue.

Benefits of the Decentralized Approach

Our approach presents several compelling advantages over centralized alternatives:

- **Reduced Costs:** No need to pay intermediary fees or commissions to ticketing agencies.
- **Security and Authenticity:** NFTs cannot be duplicated or forged, preventing fraudulent ticketing.
- **Transparency:** Anyone can verify the existence, price, and owner of a ticket on the blockchain.
- **Ownership and Control:** Users have full control of their tickets and can resell or cancel at will.
- **Automation:** Smart contracts eliminate manual work and reduce human error or manipulation.

Conclusion

The Decentralized Event Ticketing System represents a significant step forward in how we think about and manage event ticketing. By harnessing the power of blockchain and the unique properties of NFTs, the system directly addresses long-standing issues like ticket fraud, centralization, and a lack of transparency that have plagued traditional platforms.

Using the Ethereum blockchain and the ERC-721 standard, we have built a secure, trustless environment where organizers can create and manage events with ease, and users have full control over their tickets. Buyers not only own their tickets outright but can also resell or cancel them securely without relying on intermediaries or facing the risk of counterfeits.

The smart contract handles everything from ticket issuance to refunds and resales, ensuring automation, accuracy, and transparency at every step. Our extensive testing on Ethereum testnet showed that the system performs reliably across a range of real-world scenarios.

As the use of blockchain technology continues to expand, systems like this will play an important role in giving power back to users ensuring fairness, reducing costs, and making event access more secure and transparent for everyone. This project demonstrates that decentralized solutions aren't just theoretical, they are practical, scalable, and ready to reshape how we engage with live events.

Future Improvements

The system can be enhanced in several ways to improve scalability, accessibility, and compliance:

- ***Mobile Wallet Integration:*** Build a dedicated app for ticket scanning, buying, and event management.
- ***Dynamic Pricing Algorithm:*** Introduce price adjustments based on demand using oracles.
- ***Layer 2 Scaling:*** Move transactions to L2 solutions like Arbitrum or zkRollups to reduce gas fees.
- ***KYC Integration:*** Allow organizers to enforce optional identity verification for regulatory events.
- ***Event Analytics:*** Provide organizers with insights into buyer demographics, resale patterns, and attendance rates.

References

GitHub Repository:

<https://github.com/dappuniversity/tokenmaster>

YouTube Tutorial:

https://youtu.be/_H9Qppf13GI?si=dOK41cfasAl8t0bg